RAFFLES INSTITUTION
H2 Mathematics 9758
2026 Year 6 Term 3 Revision 4 (Delta)
Topic(s): Calculus II
(Integration, Applications, DE) (Solutions)

1 Find

(a) $\int \sin^{-1} 2x \, \mathrm{d}x$, [3]

(b) $\int \frac{1}{x^2 - 2x + 4} \, \mathrm{d}x$. [2]

Solution:

(i)

$$\int \sin^{-1} 2x \, \mathrm{d}x = \int 1.\sin^{-1} 2x \, \mathrm{d}x$$

$$= x\sin^{-1} 2x - \int x.\frac{2}{\sqrt{1-4x^2}} \, \mathrm{d}x$$

$$= x\sin^{-1} 2x + \frac{1}{4}\int (-8x)\left(1-4x^2\right)^{-\frac{1}{2}} \, \mathrm{d}x$$

$$= x\sin^{-1} 2x + \frac{1}{4}.\frac{\left(1-4x^2\right)^{\frac{1}{2}}}{\frac{1}{2}} + c$$

$$= x\sin^{-1} 2x + \frac{1}{2}\sqrt{1-4x^2} + c$$

$u = \sin^{-1} 2x$ $\quad \frac{\mathrm{d}v}{\mathrm{d}x} = 1$

$\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{\sqrt{1-(2x)^2}} \cdot 2$ $\quad v = x$

(ii)

$$\int \frac{1}{x^2 - 2x + 4} \, \mathrm{d}x = \int \frac{1}{(x-1)^2 + 3} \, \mathrm{d}x$$

$$= \frac{1}{\sqrt{3}} \tan^{-1}\left(\frac{x-1}{\sqrt{3}}\right) + c$$

2 **(a)** Find $\int \frac{1}{\sqrt{\left(1-x^2\right)\sin^{-1}x}}\,\mathrm{d}x$. [2]

(b) Find $\int \frac{x-3}{x^2-4x+4}\,\mathrm{d}x$. [3]

Solution:

(a)

$$\int \frac{1}{\sqrt{\left(1-x^2\right)\sin^{-1}x}}\,\mathrm{d}x = \int \frac{1}{\sqrt{\left(1-x^2\right)}}\left(\sin^{-1}x\right)^{-\frac{1}{2}}\,\mathrm{d}x$$

$$= \frac{\left(\sin^{-1}x\right)^{\frac{1}{2}}}{\frac{1}{2}} + C$$

$$= 2\sqrt{\sin^{-1}x} + C$$

For $n \neq -1$,

$$\int \mathrm{f}'(x)\left[\mathrm{f}(x)\right]^n\,\mathrm{d}x = \frac{\left[\mathrm{f}(x)\right]^{n+1}}{n+1} + c$$

$\mathrm{f}(x) = \sin^{-1}x$ and $n = -\frac{1}{2}$.

(b)

$$\int \frac{x-3}{x^2-4x+4}\,\mathrm{d}x = \int \frac{x-3}{(x-2)^2}\,\mathrm{d}x$$

$$= \int \frac{1}{x-2}\,\mathrm{d}x - \int \frac{1}{(x-2)^2}\,\mathrm{d}x$$

$$= \ln|x-2| + \frac{1}{x-2} + C$$

Alternatively,

$$\int \frac{x-3}{x^2-4x+4}\,\mathrm{d}x = \int \frac{\frac{1}{2}(2x-4)-1}{x^2-4x+4}\,\mathrm{d}x$$

$$= \frac{1}{2}\int \frac{2x-4}{x^2-4x+4}\,\mathrm{d}x - \int \frac{1}{x^2-4x+4}\,\mathrm{d}x$$

$$= \frac{1}{2}\ln\left|x^2-4x+4\right| - \int \frac{1}{(x-2)^2}\,\mathrm{d}x$$

$$= \frac{1}{2}\ln\left|(x-2)^2\right| + \frac{1}{x-2} + C$$

$$= \ln|x-2| + \frac{1}{x-2} + C$$

$$\int \frac{\mathrm{f}'(x)}{\mathrm{f}(x)}\,\mathrm{d}x = \ln|\mathrm{f}(x)| + c$$

3 **(a)** **(i)** Differentiate $\sqrt{x^2+4}$ with respect to x. [1]

(ii) Hence find $\int \frac{x^3}{\sqrt{x^2+4}}\,\mathrm{d}x$. [3]

(b) Find

(i) $\int \sin^2 mx\,\mathrm{d}x$, where $m > 0$, where m is a non-zero constant, [2]

(ii) $\int \cos^3 3x\,\mathrm{d}x$. [3]

Solution:

(a)(i)

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\sqrt{x^2+4}\right) = \frac{2x}{2\sqrt{x^2+4}} = \frac{x}{\sqrt{x^2+4}}$$

(a)(ii)

$$u = x^2 \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = \frac{x}{\sqrt{x^2+4}}$$

$$\frac{\mathrm{d}u}{\mathrm{d}x} = 2x \qquad v = \sqrt{x^2+4}$$

$$\int \frac{x^3}{\sqrt{x^2+4}}\,\mathrm{d}x$$

$$= \int x^2\left(\frac{x}{\sqrt{x^2+4}}\right)\mathrm{d}x$$

$$= x^2\sqrt{x^2+4} - \int 2x\sqrt{x^2+4}\,\mathrm{d}x$$

$$= x^2\sqrt{x^2+4} - \frac{2\left(x^2+4\right)^{\frac{3}{2}}}{3} + C$$

(b)(i)

$$\int \sin^2 mx\,\mathrm{d}x = \frac{1}{2}\int 1-\cos 2mx\,\mathrm{d}x$$

$$= \frac{1}{2}x - \frac{\sin 2mx}{4m} + C$$

(b)(ii)

$$\int \cos^3 3x\,\mathrm{d}x = \int \cos 3x\left(\cos^2 3x\right)\mathrm{d}x$$

$$= \int \cos 3x\left(1-\sin^2 3x\right)\mathrm{d}x$$

$$= \int \cos 3x - \cos 3x \sin^2 3x\,\mathrm{d}x$$

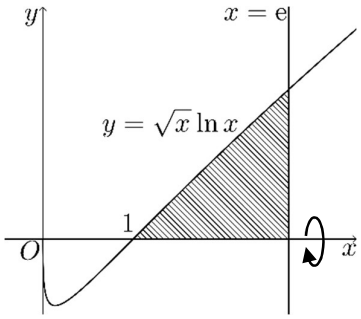
$$= \frac{\sin 3x}{3} - \frac{\sin^3 3x}{9} + C$$

4 Find the area of the region bounded by the curve $y = \sqrt{x}\ln x$, the x-axis and the line $x = \mathrm{e}$. This region is rotated completely about the x-axis, find the exact volume of the solid generated. [6]

Solution:

From GC,

Area
$= \int_1^{\mathrm{e}} \sqrt{x}\ln x\,\mathrm{d}x = 1.44$



Volume
$= \pi\int_1^{\mathrm{e}} x(\ln x)^2\,\mathrm{d}x$

$u = (\ln x)^2$ $\quad \frac{\mathrm{d}v}{\mathrm{d}x} = x$

$\frac{\mathrm{d}u}{\mathrm{d}x} = 2(\ln x)\frac{1}{x}$ $\quad v = \frac{x^2}{2}$

$= \pi\left\{\left[\frac{x^2}{2}(\ln x)^2\right]_1^{\mathrm{e}} - \int_1^{\mathrm{e}} \frac{x^2}{2}\cdot 2(\ln x)\frac{1}{x}\,\mathrm{d}x\right\}$

$= \pi\left[\frac{\mathrm{e}^2}{2} - \int_1^{\mathrm{e}} x(\ln x)\,\mathrm{d}x\right]$

$u = \ln x$ $\quad \frac{\mathrm{d}v}{\mathrm{d}x} = x$

$\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{x}$ $\quad v = \frac{x^2}{2}$

$= \pi\left\{\frac{\mathrm{e}^2}{2} - \left[\left[\frac{x^2}{2}(\ln x)\right]_1^{\mathrm{e}} - \int_1^{\mathrm{e}} \frac{x^2}{2}\cdot\frac{1}{x}\,\mathrm{d}x\right]\right\}$

$= \pi\left\{\frac{\mathrm{e}^2}{2} - \frac{\mathrm{e}^2}{2} + \left[\frac{x^2}{4}\right]_1^{\mathrm{e}}\right\}$

$= \frac{\pi}{4}(\mathrm{e}^2 - 1)$ units^3

5 Figure 1 shows a sketch of the curve $y=\mathrm{f}(x)$ and A is the region under the curve between $x=1$ and $x=3$. Yvonne and Irene use different ways to draw 4 rectangles of equal width, h, to estimate the area of A.

Figure 2 shows 4 rectangles drawn by Yvonne, with the curve intersecting each rectangle at the mid-point of its width. The x-coordinates of the mid-points are m_1, m_2, m_3 and m_4.

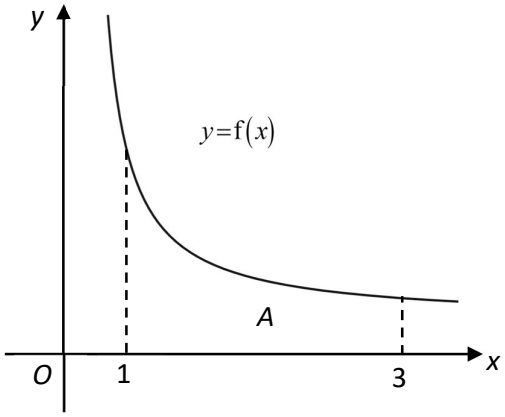


Figure 1

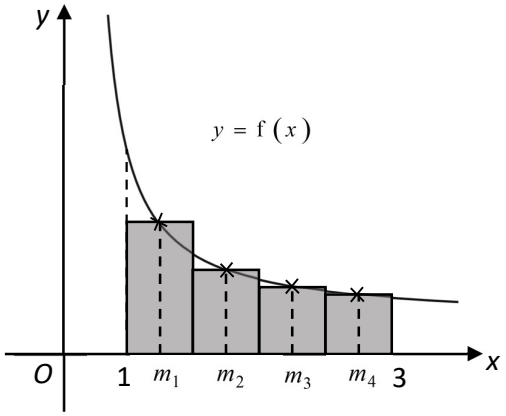


Figure 2

(a) State the values of h, m_1, m_2, m_3 and m_4. [2]

(b) The sum of area of rectangles using Yvonne's method is denoted by B. Show that $B=h\sum_{r=0}^{3}\left(\mathrm{f}\left(a+rh\right)\right)$, where the value of a is to be determined. [2]

(c) Irene finds that the sum of area of 4 rectangles that she has drawn is

$$C=h\sum_{r=0}^{3}\left(\mathrm{f}\left(1+rh\right)\right).$$

Draw these rectangles in **Figure 1**. [1]

You are now given that $\mathrm{f}(x)=\frac{1}{x}+1$.

(d) By finding the numerical values of B, C and the actual area of region A, explain how these values **and** the rectangles drawn in Figures 1 and 2 show that Yvonne's estimation of the area of A is better than Irene's. [2]

Solution:

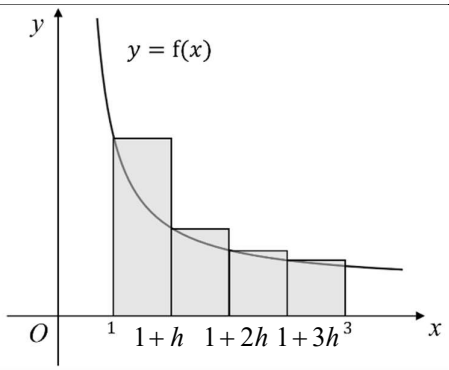
(a)	$h=\frac{3-1}{4}=0.5$ Since m_i are the mid-points of the intervals, $m_1=1.25,\ m_2=1.75,\ m_3=2.25,\ m_4=2.75.$

(b)

$$B = \mathrm{f}(m_1)\times 0.5 + \mathrm{f}(m_2)\times 0.5 + \mathrm{f}(m_3)\times 0.5 + \mathrm{f}(m_4)\times 0.5$$
$$= 0.5\times\left[\mathrm{f}(1.25) + \mathrm{f}(1.75) + \mathrm{f}(2.25) + \mathrm{f}(2.75)\right]$$
$$= 0.5\times\left[\mathrm{f}(1.25) + \mathrm{f}(1.25+1\times 0.5) + \mathrm{f}(1.25+2\times 0.5) + \mathrm{f}(1.25+3\times 0.5)\right]$$
$$= h\sum_{r=0}^{3}\mathrm{f}(1.25+rh)$$

$a = m_1 = 1.25$

(c)



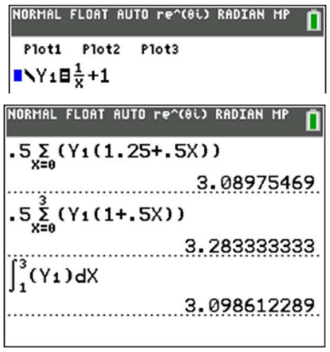
(d)

$$\mathrm{f}(x) = \frac{1}{x} + 1$$

$$A = \int_1^3 \mathrm{f}(x)\,\mathrm{d}x = 3.0986 \text{ (5sf)}$$

$$B = h\sum_{r=0}^{3}\mathrm{f}(1.25+rh) = 0.5\sum_{r=0}^{3}\mathrm{f}(1.25+0.5r) = 3.0898 \text{ (5sf) } 0$$

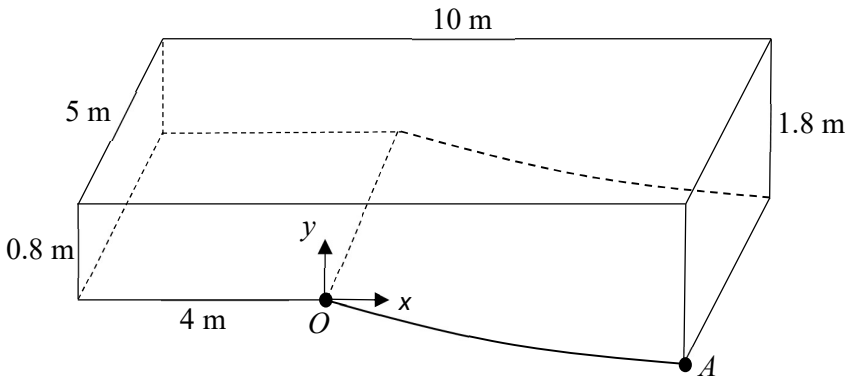
$$C = h\sum_{r=0}^{3}\mathrm{f}(1+rh) = 0.5\sum_{r=0}^{3}\mathrm{f}(1+0.5r) = 3.2833 \text{ (5sf)}$$



- The value of B is closer to the actual area of A
- Comparing Figures 1 & 2, the gaps between the curve and the top of Irene's rectangles are larger than those of Yvonne's. ***(or other equivalent explanations)***

Thus Yvonne's estimation of the actual area of A is better.

6 Tom wishes to build a pool in his backyard. The following diagram shows the dimensions of the pool he desires.



The surface of the pool is rectangular, while the floor consists of a rectangular piece and a sloped piece with a side from point O to point A as shown (with the depth of the pool gradually deepening from 0.8 m to 1.8 m). Taking O as the origin, the curve OA can be modelled by the equation $y = -\sqrt{hx+k}$, where h and k are real constants.

(i) Find the values of h and k. [2]

(ii) Hence find the volume of water required to fill the pool to the brim. [2]

Tom's wife, Jane, prefers a more artistic pool design, involving the curve C given by the equation $y = \mathrm{e}^{0.04x^2} - 3$. Her preferred pool is formed by rotating the region bounded by the curve C and the x-axis π radians about the y-axis.

(iii) What is the exact volume of water required to fill Jane's preferred pool? Give your answer in the form $(a\ln b - c)\pi$ m^3, where a, b and c are real constants to be determined. [5]

Solution:

(i)	$y = -\sqrt{hx+k}$ At $(0,0)$: $0 = -\sqrt{k} \Rightarrow k = 0$ Point A has coordinates $(6,-1)$. At $(6,-1)$: $-1 = -\sqrt{6h+0} \Rightarrow h = \frac{1}{6}$
(ii)	Volume of water required: $(10\times 5\times 0.8) + 5\times\int_0^6 \left\lvert -\sqrt{\frac{x}{6}} \right\rvert \mathrm{d}x = 60$ m^3
(iii)	$y = \mathrm{e}^{0.04x^2} - 3$

When $x = 0,\ y = 1 - 3 \Rightarrow y = -2$.

$y + 3 = \mathrm{e}^{0.04x^2}$

$\ln(y+3) = 0.04x^2$

$x^2 = 25\ln(y+3)$

Volume of water required to fill Jane's preferred pool:

$$\pi\int_{-2}^{0} x^2\,\mathrm{d}y = 25\pi\int_{-2}^{0} \ln(y+3)\,\mathrm{d}y$$

$$= 25\pi\left(\left[y\ln(y+3)\right]_{-2}^{0} - \int_{-2}^{0}\frac{y}{y+3}\,\mathrm{d}y\right)$$

$$= -25\pi\left(\int_{-2}^{0} 1 - \frac{3}{y+3}\,\mathrm{d}y\right)$$

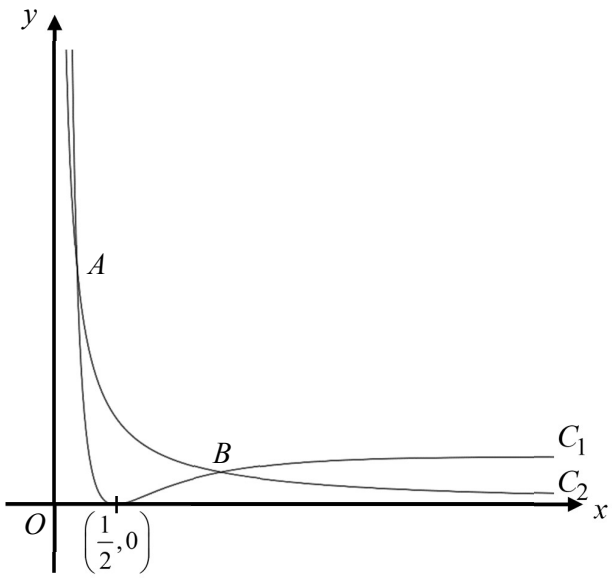
$$= -25\pi\left(\left[y - 3\ln(y+3)\right]_{-2}^{0}\right)$$

$$= -25\pi\left(-3\ln 3 - (-2)\right)$$

$$= (75\ln 3 - 50)\pi\ \mathrm{m}^3$$

$a = 75,\ b = 3,\ c = 50.$

7 The curves C_1 and C_2 have equations $y = \dfrac{(\ln 2x)^2}{x}$ and $y = \dfrac{1}{x}$ respectively. The diagram shows the parts of C_1 and C_2 for which $x > 0$. The curves C_1 and C_2 intersect at the points *A* and *B*. The curve C_1 has a turning point at $\left(\dfrac{1}{2}, 0\right)$.



(i) Find the exact *x*-coordinates of *A* and *B*. [2]

(ii) A flat blade with negligible thickness is made in the shape of the region enclosed by C_1, C_2 and the line $x = 4$. Find the exact area of the blade. [4]

(iii) A 3D model of an artefact is formed by rotating the region enclosed by C_1 and C_2 through 2π radians about the *x*-axis. Find the volume of the artefact, giving your answer correct to 3 decimal places. [2]

Solution:

(i)

$$\frac{(\ln 2x)^2}{x} = \frac{1}{x}$$

$$(\ln 2x)^2 = 1$$

$$\ln 2x = \pm 1$$

$$2x = \mathrm{e}^{\pm 1}$$

$$x = \frac{\mathrm{e}^{-1}}{2} \text{ or } \frac{\mathrm{e}}{2}$$

$$y = \frac{2}{\mathrm{e}^{-1}} \text{ or } \frac{2}{\mathrm{e}}$$

Coordinates of A and B are $\left(\frac{\mathrm{e}^{-1}}{2}, \frac{2}{\mathrm{e}^{-1}}\right)$ and $\left(\frac{\mathrm{e}}{2}, \frac{2}{\mathrm{e}}\right)$ respectively.

(ii)

Area of blade

$$= \int_{\frac{\mathrm{e}}{2}}^{4} \frac{(\ln 2x)^2}{x} - \frac{1}{x} \, \mathrm{d}x$$

$$= \left[\frac{(\ln 2x)^3}{3} - \ln x\right]_{\frac{\mathrm{e}}{2}}^{4}$$

$$= \frac{(\ln 8)^3}{3} - \ln 4 - \left(\frac{(\ln \mathrm{e})^3}{3} - \ln \frac{\mathrm{e}}{2}\right)$$

$$= \frac{(3\ln 2)^3}{3} - 2\ln 2 - \left[\frac{1}{3} - (\ln \mathrm{e} - \ln 2)\right]$$

$$= 9(\ln 2)^3 - 2\ln 2 - \frac{1}{3} + 1 - \ln 2$$

$$= 9(\ln 2)^3 - 3\ln 2 + \frac{2}{3}$$

(iii)

Volume of revolution

$$= \pi \int_{\frac{1}{2\mathrm{e}}}^{\frac{\mathrm{e}}{2}} \frac{1}{x^2} - \frac{(\ln 2x)^4}{x^2} \, \mathrm{d}x$$

$$= 11.297 \text{ units}^3 \text{ (3 d.p)}$$

8 It is given that $\left(2xy\dfrac{\mathrm{d}y}{\mathrm{d}x}+y^2\right)\cos x=\dfrac{1}{xy^2}$, where $0<x<\dfrac{\pi}{2}$ and $y\neq 0$.

(a) Using the substitution $u=xy^2$, show that the differential equation can be reduced to

$$\frac{\mathrm{d}u}{\mathrm{d}x}=\frac{\sec x}{u}.$$ [3]

(b) Hence find the general solution to the differential equation. [3]

Solution:

(a)	$u=xy^2$ $\dfrac{\mathrm{d}u}{\mathrm{d}x}=y^2+x\left(2y\dfrac{\mathrm{d}y}{\mathrm{d}x}\right)$ $\cos x\left(y^2+2xy\dfrac{\mathrm{d}y}{\mathrm{d}x}\right)=\dfrac{1}{xy^2}$ $\cos x\left(\dfrac{\mathrm{d}u}{\mathrm{d}x}\right)=\dfrac{1}{u}$ $\dfrac{\mathrm{d}u}{\mathrm{d}x}=\dfrac{\sec x}{u}$
(b)	$\int u\ \mathrm{d}u=\int \sec x\ \mathrm{d}x$ $\dfrac{1}{2}u^2=\ln\left(\sec x+\tan x\right)+C$ $\left(\because \sec x,\tan x>0 \text{ for } 0<x<\dfrac{\pi}{2}\right)$ $x^2y^4=2\ln\left(\sec x+\tan x\right)+C$

9 It is given that $2xy\dfrac{\mathrm{d}y}{\mathrm{d}x} = x^2 - y^2$ where $x > 0,\ y < 0$. Using the substitution $u = \dfrac{y}{x}$, show that the differential equation can be transformed to $\dfrac{\mathrm{d}u}{\mathrm{d}x} = \dfrac{1-3u^2}{2ux}$. Hence find the general solution of y in terms of x. [6]

Solution:

$u = \dfrac{y}{x} \Rightarrow y = ux \Rightarrow \dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{\mathrm{d}u}{\mathrm{d}x}x + u$ ... (1)

$2xy\dfrac{\mathrm{d}y}{\mathrm{d}x} = x^2 - y^2$

$\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{x^2 - y^2}{2xy} = \dfrac{x^2 - u^2x^2}{2ux^2} = \dfrac{1-u^2}{2u}$ ... (2)

Equate (1) with (2):

$\dfrac{\mathrm{d}u}{\mathrm{d}x}x + u = \dfrac{1-u^2}{2u}$

$\dfrac{\mathrm{d}u}{\mathrm{d}x}x = \dfrac{1-u^2}{2u} - u = \dfrac{1-3u^2}{2u}$

$\dfrac{\mathrm{d}u}{\mathrm{d}x} = \dfrac{1-3u^2}{2ux}$ (shown)

$\displaystyle\int \frac{2u}{1-3u^2}\,\mathrm{d}u = \int \frac{1}{x}\,\mathrm{d}x$

$\displaystyle -\frac{1}{3}\int \frac{-6u}{1-3u^2}\,\mathrm{d}u = \int \frac{1}{x}\,\mathrm{d}x$

$-\dfrac{1}{3}\ln\left|1-3u^2\right| = \ln x + c \quad (\because x > 0)$

$\ln\left|1-3u^2\right| = -3\ln x - 3c$

$1-3u^2 = \pm \mathrm{e}^{-3c}\cdot \mathrm{e}^{\ln\frac{1}{x^3}} = \dfrac{A}{x^3}$ (where $A = \pm\mathrm{e}^{-3c}$)

$3u^2 = 1 - \dfrac{A}{x^3}$

$\left(\dfrac{y}{x}\right)^2 = \dfrac{1}{3}\left(1 - \dfrac{A}{x^3}\right)$

$y = -\sqrt{\dfrac{x^2}{3}\left(1 - \dfrac{A}{x^3}\right)} = -x\sqrt{\dfrac{1}{3}\left(1 - \dfrac{A}{x^3}\right)} \quad (\because y < 0)$

10 The game designer for a video game is running a simulation of a particular stage where the Hero is trapped underground and is trying to make his way back up to the ground level. The Hero can only move vertically upwards. At time t minutes, his distance from the starting point is x units.

The Hero starts to move vertically upwards from the starting point at a rate inversely proportional to x. However, he encounters traps and monsters along the way, which will push him vertically downwards at a rate proportional to x.

(i) Show that the Hero's movement can be modelled by the differential equation

$$\frac{\mathrm{d}x}{\mathrm{d}t} = \frac{a - kx^2}{x}$$, where a and k are positive constants. [1]

During the simulation, the game designer sees k as the "difficulty factor" and sets $a = 8$.

(ii) Show that

$$x = \sqrt{\frac{8 - 8\mathrm{e}^{-2kt}}{k}}\ .$$ [5]

(iii) If the simulation is left to run for a long time, deduce the eventual position of the Hero. [2]

(iv) Hence, sketch the graph of x against t. [2]

(v) If the ground level is supposed to be 4 units above the Hero's starting point, what is the range of values that the game designer should set for the "difficulty factor" such that the Hero is able to make it to the ground level and complete the stage? [2]

Solution:

(i)

$\frac{\mathrm{d}x}{\mathrm{d}t} = (\text{rate of increase due to upward movement})$

$\quad -(\text{rate of decrease due to encounters with traps and monsters})$

$= \frac{a}{x} - bx = \frac{a - bx^2}{x}$ where a and b are positive constants

$\therefore\ \frac{\mathrm{d}x}{\mathrm{d}t} = \frac{a - kx^2}{x}$ where $k = b$ (shown)

(ii)

$$\frac{\mathrm{d}x}{\mathrm{d}t} = \frac{8-kx^2}{x}$$

$$\int \frac{x}{8-kx^2}\,\mathrm{d}x = \int 1\,\mathrm{d}t$$

$$-\frac{1}{2k}\ln\left|8-kx^2\right| = t + C$$

$$\ln\left|8-kx^2\right| = -2kt - 2kC$$

$$\left|8-kx^2\right| = \mathrm{e}^{-2kt-2kC}$$

$$8-kx^2 = A\mathrm{e}^{-2kt}, \quad A = \pm\mathrm{e}^{-2kC}$$

When $t=0,\ x=0,$

$$8-0 = A\mathrm{e}^{0}$$

$$A = 8$$

$$\therefore 8-kx^2 = 8\mathrm{e}^{-2kt}$$

$$kx^2 = 8-8\mathrm{e}^{-2kt}$$

$$x = \sqrt{\frac{8-8\mathrm{e}^{-2kt}}{k}} \quad \text{or} \quad -\sqrt{\frac{8-8\mathrm{e}^{-2kt}}{k}} \quad (\text{reject } \because x \geq 0)\ (\text{shown})$$

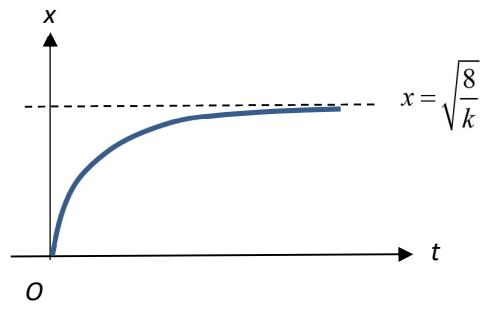
(iii)

When $t \to \infty,\ \mathrm{e}^{-2kt} \to 0,$

hence $x \to \sqrt{\frac{8-8(0)}{k}} = \sqrt{\frac{8}{k}}$ units

The Hero eventually will be stranded at a point $\sqrt{\frac{8}{k}}$ units from the starting point.

(iv)



(v)

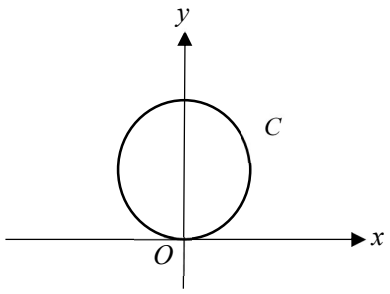
Completion point must occur before point of being stranded, therefore

$$\sqrt{\frac{8}{k}} > 4$$

$$\frac{8}{k} > 16$$

Hence $0 < k < 0.5$

11 A spherical container of radius 5 m is formed by rotating the following circle C about the y-axis.



The container has negligible thickness, and the circle C passes through the origin O.

(a) State a cartesian equation of C. [1]

Initially the spherical container is completely filled with water. Two engineers are calculating the time needed for the container to be completely drained from a small circular hole at the bottom. The volume of water in the container at time t seconds is denoted by V m^3.

[The volume of a sphere of radius r is $\frac{4}{3}\pi r^3$.]

(b) The first engineer proposes that the rate of change of V with respect to t is a constant k.

(i) Write down a differential equation relating V, t and k. [1]

(ii) Determine, with justification, the sign of k. [1]

(iii) Find V in terms of t and k, leaving your answer in exact form. [2]

(c) The second engineer argues that the rate at which water flows out from the hole will be at its greatest in the beginning, and decreases as the depth of water in the container decreases. He suggests using Torricelli's law, which says that

$$\frac{\mathrm{d}V}{\mathrm{d}t} = -\alpha\sqrt{20h},$$

where α m^2 is the area of the circular hole at the bottom of the container, and h m is the depth of water in the container at time t seconds. The radius of the hole at the bottom of the container is found to be constant at 1 cm.

(i) Show that $\left(10h^{\frac{1}{2}} - h^{\frac{3}{2}}\right)\frac{\mathrm{d}h}{\mathrm{d}t} = -\frac{\sqrt{5}}{5000}$. [4]

(ii) Hence find the numerical value of t when the container is completely drained. [4]

Solution:

(a)

Cartesian equation of C: $x^2+(y-5)^2=25$

(b)(i)

$\dfrac{\mathrm{d}V}{\mathrm{d}t}=k$

(b)(ii)

As the volume of water in the container is decreasing with time, $k<0$.

(b)(iii)

$\dfrac{\mathrm{d}V}{\mathrm{d}t}=k$

Integrate w.r.t t,

$V=kt+C$, where $C\in\mathbb{R}$

When $t=0$,

$\dfrac{4}{3}\pi(5)^3=k(0)+C\Rightarrow C=\dfrac{500\pi}{3}$

$\therefore V=kt+\dfrac{500\pi}{3}$

(c)(i)

Area of hole, $\alpha=\pi\left(\dfrac{1}{100}\right)^2=\dfrac{\pi}{10000}\text{m}^2$

So $\dfrac{\mathrm{d}V}{\mathrm{d}t}=-\dfrac{\pi}{10000}\sqrt{20h}=-\dfrac{\sqrt{5}\pi}{5000}\sqrt{h}$

$V=\pi\int_0^h x^2\,\mathrm{d}y$

$=\pi\int_0^h 25-(y-5)^2\,\mathrm{d}y$

$=\pi\int_0^h 10y-y^2\,\mathrm{d}y$

$=\pi\left[5y^2-\dfrac{y^3}{3}\right]_0^h$

$=\pi\left[5h^2-\dfrac{h^3}{3}\right]$

$\dfrac{\mathrm{d}V}{\mathrm{d}h}=\pi(10h-h^2)$

Recall: $\dfrac{\mathrm{d}V}{\mathrm{d}t}=\dfrac{\mathrm{d}V}{\mathrm{d}h}\times\dfrac{\mathrm{d}h}{\mathrm{d}t}$

So we need to find $\dfrac{\mathrm{d}V}{\mathrm{d}h}$

Since $\dfrac{\mathrm{d}V}{\mathrm{d}h}\times\dfrac{\mathrm{d}h}{\mathrm{d}t}=\dfrac{\mathrm{d}V}{\mathrm{d}t}$, we have

$\pi(10h-h^2)\dfrac{\mathrm{d}h}{\mathrm{d}t}=-\dfrac{\sqrt{5}\pi}{5000}\sqrt{h}$

$\left(10h^{\frac{1}{2}}-h^{\frac{3}{2}}\right)\dfrac{\mathrm{d}h}{\mathrm{d}t}=-\dfrac{\sqrt{5}}{5000}$ (shown)

(c)(ii)

$$\left(10h^{\frac{1}{2}} - h^{\frac{3}{2}}\right)\frac{\mathrm{d}h}{\mathrm{d}t} = -\frac{\sqrt{5}}{5000}$$

Integrate w.r.t t,

$$\int 10h^{\frac{1}{2}} - h^{\frac{3}{2}}\ \mathrm{d}h = \int -\frac{\sqrt{5}}{5000}\ \mathrm{d}t$$

$$10\left(\frac{2}{3}h^{\frac{3}{2}}\right) - \frac{2}{5}h^{\frac{5}{2}} = -\frac{\sqrt{5}}{5000}t + B$$

When $t = 0$, $h = 10$:

$$B = \frac{20}{3}(10)^{\frac{3}{2}} - \frac{2}{5}(10)^{\frac{5}{2}} = 84.327 \text{ (5 s.f.)}$$

When $h = 0$:

$$\frac{\sqrt{5}}{5000}t = 84.327$$

$$t = 188561$$

$$= 189000 \text{ seconds (3 s.f.)}$$